# symcoder round-trip demo

electrons=(a,b,c), muons=(p,q)

## 1 Setup

**Particle types**: electrons $= \{a, b, c\}$, muons $= \{p, q\}$
**Group G**: $G = S_{\text{electrons}} \times S_{\text{muons}} = S_3 \times S_2$, order $= 12$

Operations:
$|\mathbf{x}|$ (rank 1, symmetric)
$\mathbf{x} \cdot \mathbf{y}$ (rank 2, symmetric)
$\mathbf{x}_T \cdot \mathbf{y}_T$ (rank 2, symmetric)
$\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (rank 3, antisymmetric)

**Event vectors (3-D):**
**a**: $\mathbf{a} = (1.20,\ 0.30,\ 0.50)$

**b**: $\mathbf{b} = (0.40,\ 1.10,\ 0.20)$

**c**: $\mathbf{c} = (-0.50,\ 1.20,\ -0.70)$

**p**: $\mathbf{p} = (0.00,\ 0.00,\ 1.00)$

**q**: $\mathbf{q} = (0.00,\ 0.00,\ -1.00)$

## 2 Phase 1 Encoding (one orbit per FlavouredOperator in repS)

11 FlavouredOperators in repS

### 2.1 FlavouredOperator: $|\mathbf{x}|$ flavour $= (1, 0)$

**Canonical representative**: $|\mathbf{a}|$
**Encoder**: `SortEncoder` — stores eval of full orbit

**Orbit atoms**: $|\mathbf{a}|$, $|\mathbf{b}|$, $|\mathbf{c}|$
**Eval values**: +1.3342, +1.1874, +1.4765
**Encoded [3 reals]**: +1.1874, +1.3342, +1.4765

### 2.2 FlavouredOperator: $|\mathbf{x}|$ flavour $= (0, 1)$

**Canonical representative**: $|\mathbf{p}|$
**Encoder**: `SortEncoder` — stores eval of full orbit

**Orbit atoms**: $|\mathbf{p}|$, $|\mathbf{q}|$

**Eval values**: +1.0000, +1.0000
**Encoded [2 reals]**: +1.0000, +1.0000

### 2.3 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (2, 0)$

**Canonical representative**: $\mathbf{a} \cdot \mathbf{b}$
**Encoder**: `SortEncoder` — stores eval of full orbit

**Orbit atoms**: $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$, $\mathbf{b} \cdot \mathbf{c}$
**Eval values**: +0.9100, -0.5900, +0.9800
**Encoded [3 reals]**: -0.5900, +0.9100, +0.9800

### 2.4 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (1, 1)$

**Canonical representative**: $\mathbf{a} \cdot \mathbf{p}$
**Encoder**: `SortEncoder` — stores eval of full orbit

**Orbit atoms**: $\mathbf{a} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{p}$, $\mathbf{c} \cdot \mathbf{q}$
**Eval values**: +0.5000, -0.5000, +0.2000, -0.2000, -0.7000, +0.7000
**Encoded [6 reals]**: -0.7000, -0.5000, -0.2000, +0.2000, +0.5000, +0.7000

### 2.5 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (0, 2)$

**Canonical representative**: $\mathbf{p} \cdot \mathbf{q}$
**Encoder**: `SortEncoder` — stores eval of full orbit

**Orbit atoms**: $\mathbf{p} \cdot \mathbf{q}$
**Eval values**: -1.0000
**Encoded [1 reals]**: -1.0000

### 2.6 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour $= (2, 0)$

**Canonical representative**: $\mathbf{a}_T \cdot \mathbf{b}_T$
**Encoder**: `SortEncoder` — stores eval of full orbit

**Orbit atoms**: $\mathbf{a}_T \cdot \mathbf{b}_T$, $\mathbf{a}_T \cdot \mathbf{c}_T$, $\mathbf{b}_T \cdot \mathbf{c}_T$
**Eval values**: +0.8100, -0.2400, +1.1200
**Encoded [3 reals]**: -0.2400, +0.8100, +1.1200

### 2.7 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour $= (1, 1)$

**Canonical representative**: $\mathbf{a}_T \cdot \mathbf{p}_T$
**Encoder**: `SortEncoder` — stores eval of full orbit

**Orbit atoms**: $\mathbf{a}_T \cdot \mathbf{p}_T$, $\mathbf{a}_T \cdot \mathbf{q}_T$, $\mathbf{b}_T \cdot \mathbf{p}_T$, $\mathbf{b}_T \cdot \mathbf{q}_T$, $\mathbf{c}_T \cdot \mathbf{p}_T$, $\mathbf{c}_T \cdot \mathbf{q}_T$
**Eval values**: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
**Encoded [6 reals]**: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

### 2.8 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour $= (0, 2)$

**Canonical representative**: $\mathbf{p}_T \cdot \mathbf{q}_T$
**Encoder**: `SortEncoder` — stores eval of full orbit

**Orbit atoms**: $\mathbf{p}_T \cdot \mathbf{q}_T$
**Eval values**: +0.0000
**Encoded [1 reals]**: +0.0000

### 2.9 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour $= (3, 0)$

**Canonical representative**: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$
**Encoder**: `HalfSortEncoder` — stores |eval| of sign= +1 representatives

**Orbit atoms**: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$
**Eval values**: -0.6430
**Encoded [1 reals]**: +0.6430

### 2.10 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour $= (2, 1)$

**Canonical representative**: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$
**Encoder**: `HalfSortEncoder` — stores |eval| of sign= +1 representatives

**Orbit atoms**: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$
**Eval values**: +1.2000, -1.2000, +1.5900, -1.5900, +1.0300, -1.0300
**Encoded [6 reals]**: +1.0300, +1.0300, +1.2000, +1.2000, +1.5900, +1.5900

### 2.11 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour $= (1, 2)$

**Canonical representative**: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$
**Encoder**: `HalfSortEncoder` — stores |eval| of sign= +1 representatives

**Orbit atoms**: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$
**Eval values**: +0.0000, +0.0000, +0.0000
**Encoded [3 reals]**: +0.0000, +0.0000, +0.0000

## 3 Phase 2 Encoding (OverlapBlocks, complementarity-drop disabled)

### 3.1 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 2
**Encoded values**: +2.8284, +2.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ |\mathbf{q}|)$
$(|\mathbf{q}|,\ |\mathbf{p}|)$
Eval-pair multiset $(u, v)$:
(+1.0000, +1.0000)
(+1.0000, +1.0000)
PairFlavour overlap=(0, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(0, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ |\mathbf{p}|)$
$(|\mathbf{q}|,\ |\mathbf{q}|)$
Eval-pair multiset $(u, v)$:
$(+1.0000,\ +1.0000)$
$(+1.0000,\ +1.0000)$

### 3.2 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +6.0000, +7.9962, -11.5993, +39.9808, -86.3973, +32.8452, -97.7244, -61.3879, -6.6542, -76.5400, +15.8009, -14.2916
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ |\mathbf{a}|)$
$(|\mathbf{q}|,\ |\mathbf{a}|)$
$(|\mathbf{p}|,\ |\mathbf{b}|)$
$(|\mathbf{q}|,\ |\mathbf{b}|)$
$(|\mathbf{p}|,\ |\mathbf{c}|)$
$(|\mathbf{q}|,\ |\mathbf{c}|)$
Eval-pair multiset $(u, v)$:
$(+1.0000,\ +1.4765)$
$(+1.0000,\ +1.4765)$
$(+1.0000,\ +1.3342)$
$(+1.0000,\ +1.3342)$
$(+1.0000,\ +1.1874)$
$(+1.0000,\ +1.1874)$

### 3.3 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 4
**Encoded values**: +2.0000, +0.0000, +1.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \mathbf{p}_T \cdot \mathbf{q}_T)$
$(|\mathbf{q}|,\ \mathbf{p}_T \cdot \mathbf{q}_T)$
Eval-pair multiset $(u, v)$:
$(+1.0000,\ +0.0000)$
$(+1.0000,\ +0.0000)$

### 3.4 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 4
**Encoded values**: +2.0000, -2.0000, +0.0000, -2.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \mathbf{p} \cdot \mathbf{q})$
$(|\mathbf{q}|,\ \mathbf{p} \cdot \mathbf{q})$
Eval-pair multiset $(u, v)$:
$(+1.0000,\ -1.0000)$
$(+1.0000,\ -1.0000)$

### 3.5 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \mathbf{a}_T \cdot \mathbf{q}_T)$
$(|\mathbf{q}|,\ \mathbf{a}_T \cdot \mathbf{p}_T)$
$(|\mathbf{p}|,\ \mathbf{b}_T \cdot \mathbf{q}_T)$
$(|\mathbf{q}|,\ \mathbf{b}_T \cdot \mathbf{p}_T)$
$(|\mathbf{p}|,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
$(|\mathbf{q}|,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
Eval-pair multiset $(u, v)$:
(+1.0000, +0.0000)
(+1.0000, +0.0000)
(+1.0000, +0.0000)
(+1.0000, +0.0000)
(+1.0000, +0.0000)
(+1.0000, +0.0000)
PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \mathbf{a}_T \cdot \mathbf{p}_T)$
$(|\mathbf{q}|,\ \mathbf{a}_T \cdot \mathbf{q}_T)$
$(|\mathbf{p}|,\ \mathbf{b}_T \cdot \mathbf{p}_T)$
$(|\mathbf{q}|,\ \mathbf{b}_T \cdot \mathbf{q}_T)$
$(|\mathbf{p}|,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
$(|\mathbf{q}|,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
Eval-pair multiset $(u, v)$:
(+1.0000, +0.0000)
(+1.0000, +0.0000)
(+1.0000, +0.0000)
(+1.0000, +0.0000)
(+1.0000, +0.0000)
(+1.0000, +0.0000)

### 3.6 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +6.0000, +0.0000, +15.7800, +0.0000, +23.1200, +0.0000, +19.8321, +0.0000, +9.4242, +0.0000, +1.9370, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \mathbf{a} \cdot \mathbf{q})$
$(|\mathbf{q}|,\ \mathbf{a} \cdot \mathbf{p})$
$(|\mathbf{p}|,\ \mathbf{b} \cdot \mathbf{q})$
$(|\mathbf{q}|,\ \mathbf{b} \cdot \mathbf{p})$
$(|\mathbf{p}|,\ \mathbf{c} \cdot \mathbf{q})$
$(|\mathbf{q}|,\ \mathbf{c} \cdot \mathbf{p})$
Eval-pair multiset $(u, v)$:

$(+1.0000,\ -0.7000)$
$(+1.0000,\ -0.5000)$
$(+1.0000,\ -0.2000)$
$(+1.0000,\ +0.7000)$
$(+1.0000,\ +0.2000)$
$(+1.0000,\ +0.5000)$
PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +6.0000, +0.0000, +15.7800, +0.0000, +23.1200, +0.0000, +19.8321, +0.0000, +9.4242, +0.0000, +1.9370, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \mathbf{a}\cdot\mathbf{p})$
$(|\mathbf{q}|,\ \mathbf{a}\cdot\mathbf{q})$
$(|\mathbf{p}|,\ \mathbf{b}\cdot\mathbf{p})$
$(|\mathbf{q}|,\ \mathbf{b}\cdot\mathbf{q})$
$(|\mathbf{p}|,\ \mathbf{c}\cdot\mathbf{p})$
$(|\mathbf{q}|,\ \mathbf{c}\cdot\mathbf{q})$
Eval-pair multiset $(u, v)$:
$(+1.0000,\ -0.7000)$
$(+1.0000,\ -0.5000)$
$(+1.0000,\ -0.2000)$
$(+1.0000,\ +0.7000)$
$(+1.0000,\ +0.2000)$
$(+1.0000,\ +0.5000)$

### 3.7 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +6.0000, +3.3800, +11.2559, +16.9000, +5.0236, +32.7347, -8.0034, +30.6042, -10.0540, +13.5109, -3.3303, +2.1214
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \mathbf{a}_T\cdot\mathbf{b}_T)$
$(|\mathbf{q}|,\ \mathbf{a}_T\cdot\mathbf{b}_T)$
$(|\mathbf{p}|,\ \mathbf{a}_T\cdot\mathbf{c}_T)$
$(|\mathbf{q}|,\ \mathbf{a}_T\cdot\mathbf{c}_T)$
$(|\mathbf{p}|,\ \mathbf{b}_T\cdot\mathbf{c}_T)$
$(|\mathbf{q}|,\ \mathbf{b}_T\cdot\mathbf{c}_T)$
Eval-pair multiset $(u, v)$:
$(+1.0000,\ +1.1200)$
$(+1.0000,\ +1.1200)$
$(+1.0000,\ +0.8100)$
$(+1.0000,\ +0.8100)$
$(+1.0000,\ -0.2400)$
$(+1.0000,\ -0.2400)$

### 3.8 Block: $|\mathbf{x}| \times \mathbf{x}\cdot\mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +6.0000, +2.6000, +13.7566, +13.0000, +15.0264, +27.6329, +6.2214, +30.8987, -1.6099, +18.1337, -1.8384, +4.4679

Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \mathbf{a}\cdot\mathbf{b})$
$(|\mathbf{q}|,\ \mathbf{a}\cdot\mathbf{b})$
$(|\mathbf{p}|,\ \mathbf{a}\cdot\mathbf{c})$
$(|\mathbf{q}|,\ \mathbf{a}\cdot\mathbf{c})$
$(|\mathbf{p}|,\ \mathbf{b}\cdot\mathbf{c})$
$(|\mathbf{q}|,\ \mathbf{b}\cdot\mathbf{c})$
Eval-pair multiset $(u, v)$:
$(+1.0000,\ -0.5900)$
$(+1.0000,\ -0.5900)$
$(+1.0000,\ +0.9800)$
$(+1.0000,\ +0.9800)$
$(+1.0000,\ +0.9100)$
$(+1.0000,\ +0.9100)$

### 3.9 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
Eval-pair multiset $(u, v)$:
$(+1.0000,\ +0.0000)$
$(+1.0000,\ +0.0000)$
$(+1.0000,\ +0.0000)$
$(+1.0000,\ -0.0000)$
$(+1.0000,\ -0.0000)$
$(+1.0000,\ -0.0000)$

### 3.10 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +12.0000, +76.0580, +320.5800, +988.6013, +2326.8903, +4270.6178, +6142.2191, +6877.2694, +5857.9958, +3630.0616, +1483.7636, +314.7569
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
$(|\mathbf{p}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
... (4 more atom-pairs)

Eval-pair multiset $(u, v)$:
(+1.0000, +1.5900)
(+1.0000, +1.5900)
(+1.0000, +1.2000)
(+1.0000, +1.2000)
(+1.0000, +1.0300)
(+1.0000, +1.0300)
(+1.0000, −1.5900)
(+1.0000, −1.5900)
… (4 more eval-pairs)
PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +12.0000, +76.0580, +320.5800, +988.6013, +2326.8903, +4270.6178, +6142.2191, +6877.2694, +5857.9958, +3630.0616, +1483.7636, +314.7569
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
$(|\mathbf{p}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
… (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
(+1.0000, +1.5900)
(+1.0000, +1.5900)
(+1.0000, +1.2000)
(+1.0000, +1.2000)
(+1.0000, +1.0300)
(+1.0000, +1.0300)
(+1.0000, −1.5900)
(+1.0000, −1.5900)
… (4 more eval-pairs)

### 3.11 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 4
**Encoded values**: +4.0000, +6.8269, +5.6538, +1.9978
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(|\mathbf{p}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
Eval-pair multiset $(u, v)$:
(+1.0000, −0.6430)
(+1.0000, −0.6430)
(+1.0000, +0.6430)
(+1.0000, +0.6430)

### 3.12 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +11.3083, +53.3240, +134.2109, +190.1584, +143.8079, +45.3503
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ |\mathbf{b}|)$
$(|\mathbf{a}|,\ |\mathbf{c}|)$
$(|\mathbf{b}|,\ |\mathbf{a}|)$
$(|\mathbf{b}|,\ |\mathbf{c}|)$
$(|\mathbf{c}|,\ |\mathbf{a}|)$
$(|\mathbf{c}|,\ |\mathbf{b}|)$
Eval-pair multiset $(u, v)$:
(+1.4765, +1.1874)
(+1.4765, +1.3342)
(+1.3342, +1.1874)
(+1.3342, +1.4765)
(+1.1874, +1.3342)
(+1.1874, +1.4765)
PairFlavour overlap=(1, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1, 0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ |\mathbf{a}|)$
$(|\mathbf{b}|,\ |\mathbf{b}|)$
$(|\mathbf{c}|,\ |\mathbf{c}|)$
Eval-pair multiset $(u, v)$:
(+1.1874, +1.1874)
(+1.3342, +1.3342)
(+1.4765, +1.4765)

### 3.13 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +3.9981, +0.0000, +5.3073, +0.0000, +2.3391, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ \mathbf{p}_T \cdot \mathbf{q}_T)$
$(|\mathbf{b}|,\ \mathbf{p}_T \cdot \mathbf{q}_T)$
$(|\mathbf{c}|,\ \mathbf{p}_T \cdot \mathbf{q}_T)$
Eval-pair multiset $(u, v)$:
(+1.4765, +0.0000)
(+1.1874, +0.0000)
(+1.3342, +0.0000)

### 3.14 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +3.9981, -3.0000, +2.3073, -7.9962, -1.6590, -4.3073
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ \mathbf{p} \cdot \mathbf{q})$
$(|\mathbf{b}|,\ \mathbf{p} \cdot \mathbf{q})$
$(|\mathbf{c}|,\ \mathbf{p} \cdot \mathbf{q})$

Eval-pair multiset $(u, v)$:
$(+1.4765,\ -1.0000)$
$(+1.3342,\ -1.0000)$
$(+1.1874,\ -1.0000)$

### 3.15 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +15.9923, +0.0000, +117.1373, +0.0000, +519.6184, +0.0000, +1554.7706, +0.0000, +3305.7787, +0.0000, +5121.4826, +0.0000, +5825.2045, +0.0000, +4827.6985, +0.0000, +2843.1049, +0.0000, +1129.3678, +0.0000, +271.6938, +0.0000, +29.9358, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ \mathbf{b}_T \cdot \mathbf{p}_T)$
$(|\mathbf{a}|,\ \mathbf{b}_T \cdot \mathbf{q}_T)$
$(|\mathbf{a}|,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
$(|\mathbf{a}|,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
$(|\mathbf{b}|,\ \mathbf{a}_T \cdot \mathbf{p}_T)$
$(|\mathbf{b}|,\ \mathbf{a}_T \cdot \mathbf{q}_T)$
$(|\mathbf{b}|,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
$(|\mathbf{b}|,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+1.4765,\ +0.0000)$
$(+1.4765,\ +0.0000)$
$(+1.4765,\ +0.0000)$
$(+1.4765,\ +0.0000)$
$(+1.3342,\ +0.0000)$
$(+1.3342,\ +0.0000)$
$(+1.3342,\ +0.0000)$
$(+1.3342,\ +0.0000)$
... (4 more eval-pairs)
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +7.9962, +0.0000, +26.5993, +0.0000, +47.1165, +0.0000, +46.8716, +0.0000, +24.8287, +0.0000, +5.4714, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ \mathbf{a}_T \cdot \mathbf{p}_T)$
$(|\mathbf{a}|,\ \mathbf{a}_T \cdot \mathbf{q}_T)$
$(|\mathbf{b}|,\ \mathbf{b}_T \cdot \mathbf{p}_T)$
$(|\mathbf{b}|,\ \mathbf{b}_T \cdot \mathbf{q}_T)$
$(|\mathbf{c}|,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
$(|\mathbf{c}|,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
Eval-pair multiset $(u, v)$:
$(+1.4765,\ +0.0000)$
$(+1.4765,\ +0.0000)$
$(+1.3342,\ +0.0000)$
$(+1.3342,\ +0.0000)$
$(+1.1874,\ +0.0000)$
$(+1.1874,\ +0.0000)$

### 3.16 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24

**Encoded values**: +15.9923, +0.0000, +118.6973, +0.0000, +540.5384, +0.0000, +1681.8576, +0.0000, +3766.3481, +0.0000, +6224.2982, +0.0000, +7648.5883, +0.0000, +6936.4566, +0.0000, +4528.1048, +0.0000, +2020.0109, +0.0000, +553.0411, +0.0000, +70.2940, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(|\mathbf{a}|,\ \mathbf{b} \cdot \mathbf{p})$
$(|\mathbf{a}|,\ \mathbf{b} \cdot \mathbf{q})$
$(|\mathbf{a}|,\ \mathbf{c} \cdot \mathbf{p})$
$(|\mathbf{a}|,\ \mathbf{c} \cdot \mathbf{q})$
$(|\mathbf{b}|,\ \mathbf{a} \cdot \mathbf{p})$
$(|\mathbf{b}|,\ \mathbf{a} \cdot \mathbf{q})$
$(|\mathbf{b}|,\ \mathbf{c} \cdot \mathbf{p})$
$(|\mathbf{b}|,\ \mathbf{c} \cdot \mathbf{q})$
. . . (4 more atom-pairs)

Eval-pair multiset $(u, v)$:

$(+1.1874,\ -0.7000)$
$(+1.3342,\ -0.7000)$
$(+1.4765,\ -0.5000)$
$(+1.1874,\ -0.5000)$
$(+1.3342,\ +0.7000)$
$(+1.1874,\ +0.7000)$
$(+1.4765,\ +0.5000)$
$(+1.4765,\ -0.2000)$
. . . (4 more eval-pairs)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12

**Encoded values**: +7.9962, +0.0000, +27.3793, +0.0000, +51.1445, +0.0000, +54.8162, +0.0000, +31.8945, +0.0000, +7.8591, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(|\mathbf{a}|,\ \mathbf{a} \cdot \mathbf{p})$
$(|\mathbf{a}|,\ \mathbf{a} \cdot \mathbf{q})$
$(|\mathbf{b}|,\ \mathbf{b} \cdot \mathbf{p})$
$(|\mathbf{b}|,\ \mathbf{b} \cdot \mathbf{q})$
$(|\mathbf{c}|,\ \mathbf{c} \cdot \mathbf{p})$
$(|\mathbf{c}|,\ \mathbf{c} \cdot \mathbf{q})$

Eval-pair multiset $(u, v)$:

$(+1.4765,\ -0.7000)$
$(+1.4765,\ +0.7000)$
$(+1.3342,\ +0.5000)$
$(+1.1874,\ +0.2000)$
$(+1.3342,\ -0.5000)$
$(+1.1874,\ -0.2000)$

### 3.17 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6

**Encoded values**: +3.9981, +1.6900, +4.8633, +4.3515, +1.9181, +2.9918

Atom-pair orbit (multiset; ordering arbitrary):

$(|\mathbf{a}|,\ \mathbf{b}_T \cdot \mathbf{c}_T)$

($|\mathbf{b}|$, $\mathbf{a}_T \cdot \mathbf{c}_T$)
($|\mathbf{c}|$, $\mathbf{a}_T \cdot \mathbf{b}_T$)
Eval-pair multiset $(u, v)$:
(+1.3342, +1.1200)
(+1.4765, +0.8100)
(+1.1874, −0.2400)
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +7.9962, +3.3800, +22.8552, +22.6755, +26.7283, +59.6993, +4.7645, +76.6858, -14.1741, +47.4925, -8.2128, +11.0421
Atom-pair orbit (multiset; ordering arbitrary):
($|\mathbf{a}|$, $\mathbf{a}_T \cdot \mathbf{b}_T$)
($|\mathbf{a}|$, $\mathbf{a}_T \cdot \mathbf{c}_T$)
($|\mathbf{b}|$, $\mathbf{a}_T \cdot \mathbf{b}_T$)
($|\mathbf{b}|$, $\mathbf{b}_T \cdot \mathbf{c}_T$)
($|\mathbf{c}|$, $\mathbf{a}_T \cdot \mathbf{c}_T$)
($|\mathbf{c}|$, $\mathbf{b}_T \cdot \mathbf{c}_T$)
Eval-pair multiset $(u, v)$:
(+1.4765, +1.1200)
(+1.1874, +1.1200)
(+1.1874, +0.8100)
(+1.3342, +0.8100)
(+1.4765, −0.2400)
(+1.3342, −0.2400)

### 3.18 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +3.9981, +1.3000, +5.5306, +3.2470, +2.8502, +2.5238
Atom-pair orbit (multiset; ordering arbitrary):
($|\mathbf{a}|$, $\mathbf{b} \cdot \mathbf{c}$)
($|\mathbf{b}|$, $\mathbf{a} \cdot \mathbf{c}$)
($|\mathbf{c}|$, $\mathbf{a} \cdot \mathbf{b}$)
Eval-pair multiset $(u, v)$:
(+1.4765, +0.9100)
(+1.3342, +0.9800)
(+1.1874, −0.5900)
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +7.9962, +2.6000, +25.3559, +17.5430, +39.9914, +48.9122, +30.3174, +69.9165, +6.7367, +51.0380, -2.4527, +15.0917
Atom-pair orbit (multiset; ordering arbitrary):
($|\mathbf{a}|$, $\mathbf{a} \cdot \mathbf{b}$)
($|\mathbf{a}|$, $\mathbf{a} \cdot \mathbf{c}$)
($|\mathbf{b}|$, $\mathbf{a} \cdot \mathbf{b}$)
($|\mathbf{b}|$, $\mathbf{b} \cdot \mathbf{c}$)
($|\mathbf{c}|$, $\mathbf{a} \cdot \mathbf{c}$)
($|\mathbf{c}|$, $\mathbf{b} \cdot \mathbf{c}$)
Eval-pair multiset $(u, v)$:
(+1.4765, −0.5900)
(+1.3342, −0.5900)
(+1.4765, +0.9800)

(+1.3342,  +0.9100)
(+1.1874,  +0.9100)
(+1.1874,  +0.9800)

### 3.19 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=12, kind=ASSOC, encoded dim= 12
**Encoded values**: +15.9923, +117.1373, +519.6184, +1554.7706, +3305.7787, +5121.4826, +5825.2045, +4827.6985, +2843.1049, +1129.3678, +271.6938, +29.9358
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{a}|,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
(+1.4765,  +0.0000)
(+1.4765,  +0.0000)
(+1.4765,  +0.0000)
(+1.4765,  −0.0000)
(+1.3342,  −0.0000)
(+1.3342,  −0.0000)
(+1.3342,  +0.0000)
(+1.3342,  +0.0000)
. . . (4 more eval-pairs)
PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=12, kind=ASSOC, encoded dim= 6
**Encoded values**: +7.9962, +26.5993, +47.1165, +46.8716, +24.8287, +5.4714
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(|\mathbf{c}|,\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
Eval-pair multiset $(u, v)$:
(+1.4765,  +0.0000)
(+1.4765,  +0.0000)
(+1.3342,  +0.0000)
(+1.3342,  −0.0000)
(+1.1874,  −0.0000)
(+1.1874,  −0.0000)

### 3.20 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=12, kind=ASSOC, encoded dim= 12

**Encoded values**: +15.9923, +127.1953, +654.2952, +2406.8990, +6640.4046, +14054.4075, +22965.6441, +28758.7978, +26951.0762, +17999.4163, +7735.4211, +1640.2216
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(|\mathbf{a}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+1.1874,\ +1.5900)$
$(+1.1874,\ +1.5900)$
$(+1.4765,\ +1.2000)$
$(+1.4765,\ +1.2000)$
$(+1.3342,\ +1.0300)$
$(+1.3342,\ +1.0300)$
$(+1.1874,\ -1.5900)$
$(+1.1874,\ -1.5900)$
... (4 more eval-pairs)
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +15.9923, +0.0000, +127.1953, +0.0000, +653.3436, +0.0000, +2395.3663, +0.0000, +6572.5381, -0.0000, +13804.9212, -0.0000, +22339.0427, -0.0000, +27649.1661, -0.0000, +25568.2570, +0.0000, +16830.6354, -0.0000, +7126.1835, -0.0000, +1490.5721, -0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+1.4765,\ -1.5900)$
$(+1.3342,\ -1.5900)$
$(+1.4765,\ -1.0300)$
$(+1.3342,\ -1.2000)$
$(+1.1874,\ -1.2000)$
$(+1.1874,\ -1.0300)$
$(+1.4765,\ +1.5900)$
$(+1.3342,\ +1.5900)$
... (4 more eval-pairs)

### 3.21 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +7.9962, +27.8397, +53.7285, +60.6021, +37.9391, +10.3729

Atom-pair orbit (multiset; ordering arbitrary):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(|\mathbf{c}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
Eval-pair multiset $(u, v)$:
$(+1.4765,\ +0.6430)$
$(+1.3342,\ +0.6430)$
$(+1.1874,\ +0.6430)$
$(+1.4765,\ -0.6430)$
$(+1.3342,\ -0.6430)$
$(+1.1874,\ -0.6430)$

### 3.22 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(0, 2)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{p}_T \cdot \mathbf{q}_T)$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +0.0000)$

### 3.23 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 2) type=11 kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 2)$, type=`11`, kind=`ASSOC`, encoded dim= 2
**Encoded values**: +0.0000, -1.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{p} \cdot \mathbf{q})$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -1.0000)$

### 3.24 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{p}_T)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{q}_T)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{p}_T)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{q}_T)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$

(+0.0000, +0.0000)
(+0.0000, +0.0000)

### 3.25 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +0.0000, +0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, +0.0049, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{p})$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{q})$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{p})$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{q})$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{p})$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{q})$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -0.7000)$
$(+0.0000,\ -0.5000)$
$(+0.0000,\ +0.7000)$
$(+0.0000,\ +0.5000)$
$(+0.0000,\ -0.2000)$
$(+0.0000,\ +0.2000)$

### 3.26 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +0.0000, +1.6900, -0.4440, +0.0000, +0.0000, +0.2177
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T)$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +1.1200)$
$(+0.0000,\ +0.8100)$
$(+0.0000,\ -0.2400)$

### 3.27 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +0.0000, +1.3000, +0.2233, +0.0000, +0.0000, +0.5262
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{c})$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -0.5900)$
$(+0.0000,\ +0.9800)$
$(+0.0000,\ +0.9100)$

### 3.28 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 2)$, type=`12`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ -0.0000)$
$(+0.0000,\ -0.0000)$
$(+0.0000,\ -0.0000)$

### 3.29 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -1.5900)$
$(+0.0000,\ -1.5900)$
$(+0.0000,\ -1.2000)$
$(+0.0000,\ -1.2000)$
$(+0.0000,\ -1.0300)$
$(+0.0000,\ -1.0300)$
$(+0.0000,\ +1.5900)$
$(+0.0000,\ +1.5900)$
... (4 more eval-pairs)

### 3.30 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 2
**Encoded values**: +0.0000, +0.4134
Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -0.6430)$
$(+0.0000,\ +0.6430)$

### 3.31 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(0, 2)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{p} \cdot \mathbf{q})$
Eval-pair multiset $(u, v)$:
$(-1.0000,\ -1.0000)$

### 3.32 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: -6.0000, +0.0000, +15.0000, +0.0000, -20.0000, +0.0000, +15.0000, +0.0000, -6.0000, +0.0000, +1.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{p}_T)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{q}_T)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b}_T \cdot \mathbf{p}_T)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b}_T \cdot \mathbf{q}_T)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{c}_T \cdot \mathbf{p}_T)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{c}_T \cdot \mathbf{q}_T)$
Eval-pair multiset $(u, v)$:
$(-1.0000,\ +0.0000)$
$(-1.0000,\ +0.0000)$
$(-1.0000,\ +0.0000)$
$(-1.0000,\ +0.0000)$
$(-1.0000,\ +0.0000)$
$(-1.0000,\ +0.0000)$

### 3.33 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: -6.0000, +0.0000, +15.7800, +0.0000, -23.1200, +0.0000, +19.8321, +0.0000, -9.4242, +0.0000, +1.9370, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{p})$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{q})$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{p})$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{q})$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{p})$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{q})$
Eval-pair multiset $(u, v)$:
$(-1.0000,\ +0.7000)$
$(-1.0000,\ +0.5000)$

$(-1.0000,\ +0.2000)$
$(-1.0000,\ -0.7000)$
$(-1.0000,\ -0.2000)$
$(-1.0000,\ -0.5000)$

### 3.34 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=11, kind=ASSOC, encoded dim= 6
**Encoded values**: -3.0000, +1.6900, +2.5560, -3.3800, -0.5560, +1.9077
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{b}_T)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{c}_T)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b}_T \cdot \mathbf{c}_T)$
Eval-pair multiset $(u, v)$:
$(-1.0000,\ +1.1200)$
$(-1.0000,\ +0.8100)$
$(-1.0000,\ -0.2400)$

### 3.35 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=11, kind=ASSOC, encoded dim= 6
**Encoded values**: -3.0000, +1.3000, +3.2233, -2.6000, -1.2233, +1.8262
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{c})$
Eval-pair multiset $(u, v)$:
$(-1.0000,\ -0.5900)$
$(-1.0000,\ +0.9800)$
$(-1.0000,\ +0.9100)$

### 3.36 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 2)$, type=12, kind=ASSOC, encoded dim= 6
**Encoded values**: -6.0000, +15.0000, -20.0000, +15.0000, -6.0000, +1.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
Eval-pair multiset $(u, v)$:
$(-1.0000,\ +0.0000)$
$(-1.0000,\ +0.0000)$
$(-1.0000,\ +0.0000)$
$(-1.0000,\ -0.0000)$
$(-1.0000,\ -0.0000)$

$(-1.0000,\ -0.0000)$

### 3.37 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: -12.0000, +76.0580, -320.5800, +988.6013, -2326.8903, +4270.6178, -6142.2191, +6877.2694, -5857.9958, +3630.0616, -1483.7636, +314.7569
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(-1.0000,\ -1.5900)$
$(-1.0000,\ -1.5900)$
$(-1.0000,\ -1.2000)$
$(-1.0000,\ -1.2000)$
$(-1.0000,\ -1.0300)$
$(-1.0000,\ -1.0300)$
$(-1.0000,\ +1.5900)$
$(-1.0000,\ +1.5900)$
. . . (4 more eval-pairs)

### 3.38 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 2
**Encoded values**: -2.0000, +1.4134
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
Eval-pair multiset $(u, v)$:
$(-1.0000,\ -0.6430)$
$(-1.0000,\ +0.6430)$

### 3.39 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b}_T \cdot \mathbf{q}_T)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{p}_T)$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{p}_T)$

$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{q}_T)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{p}_T)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
. . . (4 more eval-pairs)
PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b}_T \cdot \mathbf{p}_T)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{q}_T)$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{p}_T)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{q}_T)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
. . . (4 more eval-pairs)
PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{q}_T)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{p}_T)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b}_T \cdot \mathbf{q}_T)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{p}_T)$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$

(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{p}_T)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{q}_T)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b}_T \cdot \mathbf{p}_T)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{q}_T)$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{c}_T \cdot \mathbf{p}_T)$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{q}_T)$
Eval-pair multiset $(u, v)$:
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)

### 3.40 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, +1.5600, +0.0000, +0.0000, +0.0000, +0.9126, +0.0000, +0.0000, +0.0000, +0.2471, +0.0000, +0.0000, +0.0000, +0.0308, +0.0000, +0.0000, +0.0000, +0.0015, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{q})$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{p})$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{q})$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{p})$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{q})$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{p})$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{q})$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{p})$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
(+0.0000, +0.7000)
(+0.0000, +0.7000)
(+0.0000, −0.7000)
(+0.0000, −0.7000)
(+0.0000, −0.5000)
(+0.0000, −0.5000)
(+0.0000, +0.5000)
(+0.0000, +0.5000)
. . . (4 more eval-pairs)
PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, +1.5600, +0.0000, +0.0000, +0.0000, +0.9126, +0.0000, +0.0000, +0.0000, +0.2471, +0.0000, +0.0000, +0.0000, +0.0308, +0.0000, +0.0000, +0.0000, +0.0015, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{p})$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{q})$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{p})$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{q})$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{p})$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{q})$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{p})$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{q})$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +0.7000)$
$(+0.0000,\ +0.7000)$
$(+0.0000,\ -0.7000)$
$(+0.0000,\ -0.7000)$
$(+0.0000,\ -0.5000)$
$(+0.0000,\ -0.5000)$
$(+0.0000,\ +0.5000)$
$(+0.0000,\ +0.5000)$
. . . (4 more eval-pairs)
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +0.0000, +0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, +0.0049, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{q})$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{p})$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{q})$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{p})$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{q})$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{p})$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -0.7000)$
$(+0.0000,\ -0.5000)$
$(+0.0000,\ +0.7000)$
$(+0.0000,\ +0.5000)$
$(+0.0000,\ -0.2000)$
$(+0.0000,\ +0.2000)$
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +0.0000, +0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, +0.0049, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{p})$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{q})$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{p})$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{q})$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{p})$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{q})$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -0.7000)$
$(+0.0000,\ -0.5000)$
$(+0.0000,\ +0.7000)$
$(+0.0000,\ +0.5000)$

$(+0.0000,\ -0.2000)$
$(+0.0000,\ +0.2000)$

### 3.41 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +3.3800, -3.7441, +0.0000, +0.0000, -1.0653, -0.5388, +0.0000, +0.0000, -0.1933, -0.0474, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T)$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T)$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T)$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +1.1200)$
$(+0.0000,\ +1.1200)$
$(+0.0000,\ +0.8100)$
$(+0.0000,\ +0.8100)$
$(+0.0000,\ -0.2400)$
$(+0.0000,\ -0.2400)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +6.7600, -18.9126, +0.0000, +0.0000, -27.4406, +20.1419, +0.0000, +0.0000, +3.9480, +4.1119, +0.0000, +0.0000, +2.2752, +0.2333, +0.0000, +0.0000, +0.3093, +0.0137, +0.0000, +0.0000, +0.0183, +0.0022, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T)$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T)$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +1.1200)$
$(+0.0000,\ +1.1200)$
$(+0.0000,\ +1.1200)$
$(+0.0000,\ +1.1200)$
$(+0.0000,\ +0.8100)$
$(+0.0000,\ +0.8100)$
$(+0.0000,\ +0.8100)$
$(+0.0000,\ +0.8100)$
. . . (4 more eval-pairs)

### 3.42 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=11, kind=ASSOC, encoded dim= 12
**Encoded values**: +0.0000, +2.6000, -1.2434, +0.0000, +0.0000, +1.6329, -1.3182, +0.0000, +0.0000, +0.2350, -0.2768, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{c})$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{c})$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{b})$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -0.5900)$
$(+0.0000,\ -0.5900)$
$(+0.0000,\ +0.9800)$
$(+0.0000,\ +0.9800)$
$(+0.0000,\ +0.9100)$
$(+0.0000,\ +0.9100)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=11, kind=ASSOC, encoded dim= 24
**Encoded values**: +0.0000, +5.2000, -9.2468, +0.0000, +0.0000, -3.1999, -9.5814, +0.0000, +0.0000, -10.4452, -1.1640, +0.0000, +0.0000, -6.3288, +1.6586, +0.0000, +0.0000, -1.5236, +0.6746, +0.0000, +0.0000, -0.1301, +0.0766, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{c})$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{c})$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -0.5900)$
$(+0.0000,\ -0.5900)$
$(+0.0000,\ -0.5900)$
$(+0.0000,\ -0.5900)$
$(+0.0000,\ +0.9800)$
$(+0.0000,\ +0.9800)$
$(+0.0000,\ +0.9800)$
$(+0.0000,\ +0.9800)$
... (4 more eval-pairs)

### 3.43 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 1)$, type=11, kind=ASSOC, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ -0.0000)$
$(+0.0000,\ -0.0000)$
$(+0.0000,\ -0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
... (4 more eval-pairs)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ +0.0000)$
$(+0.0000,\ -0.0000)$
$(+0.0000,\ -0.0000)$
$(+0.0000,\ -0.0000)$

### 3.44 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$

$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -1.5900)$
$(+0.0000,\ -1.5900)$
$(+0.0000,\ -1.2000)$
$(+0.0000,\ -1.2000)$
$(+0.0000,\ -1.0300)$
$(+0.0000,\ -1.0300)$
$(+0.0000,\ +1.5900)$
$(+0.0000,\ +1.5900)$
. . . (4 more eval-pairs)
PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -1.5900)$
$(+0.0000,\ -1.5900)$
$(+0.0000,\ -1.2000)$
$(+0.0000,\ -1.2000)$
$(+0.0000,\ -1.0300)$
$(+0.0000,\ -1.0300)$
$(+0.0000,\ +1.5900)$
$(+0.0000,\ +1.5900)$
. . . (4 more eval-pairs)
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, +10.0580, +0.0000, +0.0000, +0.0000, +40.9913, +0.0000, +0.0000, +0.0000, +86.6819, +0.0000, +0.0000, +0.0000, +100.4717, +0.0000, +0.0000, +0.0000, +60.6378, +0.0000, +0.0000, +0.0000, +14.9163, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -1.5900)$

(+0.0000, −1.5900)
(+0.0000, −1.2000)
(+0.0000, −1.2000)
(+0.0000, −1.0300)
(+0.0000, −1.0300)
(+0.0000, +1.5900)
(+0.0000, +1.5900)
... (4 more eval-pairs)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, +10.0580, +0.0000, +0.0000, +0.0000, +40.9913, +0.0000, +0.0000, +0.0000, +86.6819, +0.0000, +0.0000, +0.0000, +100.4717, +0.0000, +0.0000, +0.0000, +60.6378, +0.0000, +0.0000, +0.0000, +14.9163, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
(+0.0000, −1.5900)
(+0.0000, −1.5900)
(+0.0000, −1.2000)
(+0.0000, −1.2000)
(+0.0000, −1.0300)
(+0.0000, −1.0300)
(+0.0000, +1.5900)
(+0.0000, +1.5900)
... (4 more eval-pairs)

### 3.45 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +2.4807, +0.0000, +2.5641, +0.0000, +1.4135, +0.0000, +0.4383, +0.0000, +0.0725, +0.0000, +0.0050
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
(+0.0000, −0.6430)
(+0.0000, −0.6430)

$(+0.0000,\ -0.6430)$
$(+0.0000,\ -0.6430)$
$(+0.0000,\ -0.6430)$
$(+0.0000,\ -0.6430)$
$(+0.0000,\ +0.6430)$
$(+0.0000,\ +0.6430)$
... (4 more eval-pairs)

### 3.46 Block: $\mathbf{x \cdot y \times x \cdot y}$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, -1.5600, +0.0000, +0.9126, -0.0000, -0.2198, +0.0000, +0.0095, +0.0000, +0.0027, +0.0000, +0.0129
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a \cdot p},\ \mathbf{b \cdot q})$
$(\mathbf{a \cdot q},\ \mathbf{b \cdot p})$
$(\mathbf{a \cdot p},\ \mathbf{c \cdot q})$
$(\mathbf{a \cdot q},\ \mathbf{c \cdot p})$
$(\mathbf{b \cdot p},\ \mathbf{a \cdot q})$
$(\mathbf{b \cdot q},\ \mathbf{a \cdot p})$
$(\mathbf{b \cdot p},\ \mathbf{c \cdot q})$
$(\mathbf{b \cdot q},\ \mathbf{c \cdot p})$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.5000,\ +0.7000)$
$(+0.7000,\ +0.5000)$
$(+0.7000,\ +0.2000)$
$(+0.2000,\ +0.7000)$
$(+0.5000,\ -0.2000)$
$(+0.2000,\ -0.5000)$
$(-0.2000,\ -0.7000)$
$(-0.5000,\ -0.7000)$
... (4 more eval-pairs)
PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +1.5600, -0.0000, +0.9126, -0.0000, +0.2198, -0.0000, +0.0095, -0.0000, -0.0027, +0.0000, +0.0129
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a \cdot p},\ \mathbf{b \cdot p})$
$(\mathbf{a \cdot q},\ \mathbf{b \cdot q})$
$(\mathbf{a \cdot p},\ \mathbf{c \cdot p})$
$(\mathbf{a \cdot q},\ \mathbf{c \cdot q})$
$(\mathbf{b \cdot p},\ \mathbf{a \cdot p})$
$(\mathbf{b \cdot q},\ \mathbf{a \cdot q})$
$(\mathbf{b \cdot p},\ \mathbf{c \cdot p})$
$(\mathbf{b \cdot q},\ \mathbf{c \cdot q})$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.7000,\ -0.5000)$
$(+0.5000,\ -0.7000)$
$(+0.2000,\ -0.7000)$
$(+0.7000,\ -0.2000)$

$(-0.2000,\ -0.5000)$
$(-0.5000,\ -0.2000)$
$(-0.7000,\ +0.2000)$
$(-0.7000,\ +0.5000)$
… (4 more eval-pairs)
PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +0.0000, +1.5600, +0.0000, +0.6084, +0.0000, +0.0392
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{q})$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{p})$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{b}\cdot\mathbf{q})$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{b}\cdot\mathbf{p})$
$(\mathbf{c}\cdot\mathbf{p},\ \mathbf{c}\cdot\mathbf{q})$
$(\mathbf{c}\cdot\mathbf{q},\ \mathbf{c}\cdot\mathbf{p})$
Eval-pair multiset $(u, v)$:
$(+0.7000,\ -0.7000)$
$(+0.5000,\ -0.5000)$
$(-0.7000,\ +0.7000)$
$(-0.5000,\ +0.5000)$
$(+0.2000,\ -0.2000)$
$(-0.2000,\ +0.2000)$
PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{p})$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{q})$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{b}\cdot\mathbf{p})$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{b}\cdot\mathbf{q})$
$(\mathbf{c}\cdot\mathbf{p},\ \mathbf{c}\cdot\mathbf{p})$
$(\mathbf{c}\cdot\mathbf{q},\ \mathbf{c}\cdot\mathbf{q})$
Eval-pair multiset $(u, v)$:
$(-0.7000,\ -0.7000)$
$(-0.5000,\ -0.5000)$
$(-0.2000,\ -0.2000)$
$(+0.2000,\ +0.2000)$
$(+0.5000,\ +0.5000)$
$(+0.7000,\ +0.7000)$

### 3.47 Block: $\mathbf{x}\cdot\mathbf{y}\times\mathbf{x}_T\cdot\mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +3.3800, -4.5241, -0.0000, +0.0000, -2.3671, -0.0651, +0.0000, -0.0000, -0.3392, -0.1683, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{b}_T\cdot\mathbf{c}_T)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{b}_T\cdot\mathbf{c}_T)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{a}_T\cdot\mathbf{c}_T)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{a}_T\cdot\mathbf{c}_T)$
$(\mathbf{c}\cdot\mathbf{p},\ \mathbf{a}_T\cdot\mathbf{b}_T)$
$(\mathbf{c}\cdot\mathbf{q},\ \mathbf{a}_T\cdot\mathbf{b}_T)$
Eval-pair multiset $(u, v)$:

$(+0.5000,\ +1.1200)$
$(+0.7000,\ +0.8100)$
$(-0.5000,\ +1.1200)$
$(-0.7000,\ +0.8100)$
$(+0.2000,\ -0.2400)$
$(-0.2000,\ -0.2400)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +6.7600, -20.4726, +0.0000, -0.0000, -36.6844, +44.4512, -0.0000, +0.0000, +41.4253, -33.6972, -0.0000, +0.0000, -24.2761, +14.0721, +0.0000, -0.0000, +6.4766, -2.8173, -0.0000, +0.0000, -1.1257, +0.2399, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{a}_T\cdot\mathbf{b}_T)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{a}_T\cdot\mathbf{b}_T)$
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{a}_T\cdot\mathbf{c}_T)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{a}_T\cdot\mathbf{c}_T)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{a}_T\cdot\mathbf{b}_T)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{a}_T\cdot\mathbf{b}_T)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{b}_T\cdot\mathbf{c}_T)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{b}_T\cdot\mathbf{c}_T)$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(-0.7000,\ -0.2400)$
$(-0.7000,\ +1.1200)$
$(-0.5000,\ -0.2400)$
$(+0.7000,\ -0.2400)$
$(+0.5000,\ -0.2400)$
$(+0.7000,\ +1.1200)$
$(-0.5000,\ +0.8100)$
$(-0.2000,\ +1.1200)$
... (4 more eval-pairs)

### 3.48 Block: $\mathbf{x}\cdot\mathbf{y}\times\mathbf{x}\cdot\mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +2.6000, -2.0234, +0.0000, +0.0000, +0.9395, -1.6868, +0.0000, +0.0000, -0.0250, -0.6192, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{b}\cdot\mathbf{c})$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{b}\cdot\mathbf{c})$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{c})$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{c})$
$(\mathbf{c}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{b})$
$(\mathbf{c}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{b})$
Eval-pair multiset $(u, v)$:
$(-0.7000,\ +0.9100)$
$(-0.5000,\ +0.9800)$
$(+0.7000,\ +0.9100)$
$(+0.5000,\ +0.9800)$
$(-0.2000,\ -0.5900)$
$(+0.2000,\ -0.5900)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`,

kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +5.2000, -10.8068, -0.0000, +0.0000, -10.6185, +5.1708, +0.0000, -0.0000, +5.4759, -11.3264, +0.0000, +0.0000, -10.0996, +2.7457, +0.0000, +0.0000, +0.6239, -2.8179, +0.0000, -0.0000, -2.5284, +0.6807, -0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{b} \cdot \mathbf{c})$
$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{c})$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(-0.7000,\ -0.5900)$
$(-0.5000,\ -0.5900)$
$(+0.7000,\ -0.5900)$
$(+0.5000,\ -0.5900)$
$(-0.7000,\ +0.9800)$
$(+0.7000,\ +0.9800)$
$(-0.5000,\ +0.9100)$
$(+0.5000,\ +0.9100)$
... (4 more eval-pairs)

### 3.49 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, -1.5600, +0.0000, +0.0000, +0.0000, +0.9126, +0.0000, +0.0000, +0.0000, -0.2471, +0.0000, -0.0000, +0.0000, +0.0308, +0.0000, +0.0000, +0.0000, -0.0015, +0.0000, -0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a} \cdot \mathbf{p},\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a} \cdot \mathbf{q},\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a} \cdot \mathbf{p},\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a} \cdot \mathbf{q},\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{b} \cdot \mathbf{p},\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{b} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{b} \cdot \mathbf{p},\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{b} \cdot \mathbf{q},\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.7000,\ +0.0000)$
$(+0.7000,\ +0.0000)$
$(-0.7000,\ +0.0000)$
$(-0.7000,\ -0.0000)$
$(-0.5000,\ -0.0000)$
$(-0.5000,\ -0.0000)$
$(+0.5000,\ +0.0000)$
$(+0.5000,\ +0.0000)$
... (4 more eval-pairs)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 1)$, type=`11`,

kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +0.0000, -0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, -0.0049, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))$
$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
$(\mathbf{c}\cdot\mathbf{q},\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
Eval-pair multiset $(u, v)$:
$(-0.7000,\ +0.0000)$
$(-0.5000,\ +0.0000)$
$(+0.7000,\ +0.0000)$
$(+0.5000,\ -0.0000)$
$(-0.2000,\ -0.0000)$
$(+0.2000,\ -0.0000)$

### 3.50 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +8.4980, +0.0000, +32.6451, +0.0000, +74.9539, +0.0000, +108.9323, +0.0000, +93.2282, +0.0000, +42.2161
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\mathbf{c}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{p},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.2000,\ -1.5900)$
$(-0.2000,\ -1.5900)$
$(+0.7000,\ -1.2000)$
$(+0.5000,\ -1.0300)$
$(-0.7000,\ -1.2000)$
$(-0.5000,\ -1.0300)$
$(+0.7000,\ +1.2000)$
$(+0.2000,\ +1.5900)$
... (4 more eval-pairs)
PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +8.4980, +0.0000, +32.6451, +0.0000, +74.9539, +0.0000, +108.9323, +0.0000, +93.2282, +0.0000, +42.2161
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$

$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{c}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{p},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.2000,\ -1.5900)$
$(-0.2000,\ -1.5900)$
$(+0.7000,\ -1.2000)$
$(+0.5000,\ -1.0300)$
$(-0.7000,\ -1.2000)$
$(-0.5000,\ -1.0300)$
$(+0.7000,\ +1.2000)$
$(+0.2000,\ +1.5900)$
... (4 more eval-pairs)
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, +8.4980, +6.3900, -0.0000, -0.0000, +14.3494, +44.1434, -0.0000, +0.0000, -28.3068, +104.9126, -0.0000, -0.0000, -107.5651, +100.7828, +0.0000, -0.0000, -108.4640, +22.6706, +0.0000, -0.0000, -33.8292, -11.7340
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(\mathbf{b}\cdot\mathbf{p},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.7000,\ -1.5900)$
$(+0.5000,\ -1.5900)$
$(-0.2000,\ -1.2000)$
$(+0.7000,\ -1.0300)$
$(+0.5000,\ -1.2000)$
$(+0.2000,\ -1.0300)$
$(-0.7000,\ +1.5900)$
$(-0.5000,\ +1.5900)$
... (4 more eval-pairs)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, +8.4980, -6.3900, +0.0000, +0.0000, +14.3494, -44.1434, +0.0000, -0.0000, -28.3068, -104.9126, +0.0000, +0.0000, -107.5651, -100.7828, +0.0000, +0.0000, -108.4640, -22.6706, +0.0000, -0.0000, -33.8292, +11.7340
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{p},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$

$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.7000,\ +1.5900)$
$(+0.5000,\ +1.5900)$
$(-0.2000,\ +1.2000)$
$(+0.7000,\ +1.0300)$
$(+0.5000,\ +1.2000)$
$(+0.2000,\ +1.0300)$
$(-0.7000,\ -1.5900)$
$(-0.5000,\ -1.5900)$
... (4 more eval-pairs)

### 3.51 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +0.9207, -0.0000, +1.5418, -0.0000, +1.1362, -0.0000, +0.7563, -0.0000, +0.3093, -0.0000, +0.0739
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{c}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{a}\cdot\mathbf{p},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.7000,\ -0.6430)$
$(-0.7000,\ -0.6430)$
$(+0.5000,\ -0.6430)$
$(-0.5000,\ -0.6430)$
$(+0.2000,\ -0.6430)$
$(-0.2000,\ -0.6430)$
$(+0.7000,\ +0.6430)$
$(-0.7000,\ +0.6430)$
... (4 more eval-pairs)

### 3.52 Block: $\mathbf{x}_T\cdot\mathbf{y}_T\times\mathbf{x}_T\cdot\mathbf{y}_T$ (block)

PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +4.7800, +10.5364, +14.1307, +12.0340, +6.2732, +1.7889
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T\cdot\mathbf{b}_T,\ \mathbf{a}_T\cdot\mathbf{c}_T)$
$(\mathbf{a}_T\cdot\mathbf{c}_T,\ \mathbf{a}_T\cdot\mathbf{b}_T)$
$(\mathbf{a}_T\cdot\mathbf{b}_T,\ \mathbf{b}_T\cdot\mathbf{c}_T)$
$(\mathbf{b}_T\cdot\mathbf{c}_T,\ \mathbf{a}_T\cdot\mathbf{b}_T)$
$(\mathbf{a}_T\cdot\mathbf{c}_T,\ \mathbf{b}_T\cdot\mathbf{c}_T)$
$(\mathbf{b}_T\cdot\mathbf{c}_T,\ \mathbf{a}_T\cdot\mathbf{c}_T)$

Eval-pair multiset $(u, v)$:
$(-0.2400,\ +1.1200)$
$(-0.2400,\ +0.8100)$
$(+0.8100,\ +1.1200)$
$(+1.1200,\ +0.8100)$
$(+1.1200,\ -0.2400)$
$(+0.8100,\ -0.2400)$
PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2, 0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T)$
Eval-pair multiset $(u, v)$:
$(-0.2400,\ -0.2400)$
$(+0.8100,\ +0.8100)$
$(+1.1200,\ +1.1200)$

### 3.53 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +3.3800, +2.6000, +2.5007, +8.5673, -4.5116, +11.3877, -9.5180, +5.4343, -6.3584, -0.9672, -1.2926, -1.7926
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \mathbf{b} \cdot \mathbf{c})$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \mathbf{b} \cdot \mathbf{c})$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \mathbf{a} \cdot \mathbf{c})$
Eval-pair multiset $(u, v)$:
$(+1.1200,\ -0.5900)$
$(+0.8100,\ -0.5900)$
$(+1.1200,\ +0.9100)$
$(+0.8100,\ +0.9800)$
$(-0.2400,\ +0.9800)$
$(-0.2400,\ +0.9100)$
PairFlavour overlap=(2, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(2, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +1.6900, +1.3000, +0.6673, +0.2207, +1.0660, -0.4442
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \mathbf{b} \cdot \mathbf{c})$
Eval-pair multiset $(u, v)$:
$(+1.1200,\ +0.9800)$
$(+0.8100,\ +0.9100)$
$(-0.2400,\ -0.5900)$

### 3.54 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +3.3800, +3.7441, +1.0653, -0.5388, -0.1933, +0.0474
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
Eval-pair multiset $(u, v)$:
$(+1.1200,\ +0.0000)$
$(+1.1200,\ +0.0000)$
$(+0.8100,\ +0.0000)$
$(+0.8100,\ -0.0000)$
$(-0.2400,\ -0.0000)$
$(-0.2400,\ -0.0000)$
PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +6.7600, +18.9126, +27.4406, +20.1419, +3.9480, -4.1119, -2.2752, +0.2333, +0.3093, -0.0137, -0.0183, +0.0022
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+1.1200,\ +0.0000)$
$(+1.1200,\ +0.0000)$
$(+1.1200,\ +0.0000)$
$(+1.1200,\ -0.0000)$
$(+0.8100,\ -0.0000)$
$(+0.8100,\ -0.0000)$
$(+0.8100,\ +0.0000)$
$(+0.8100,\ +0.0000)$
... (4 more eval-pairs)

### 3.55 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +6.7600, +0.0000, +28.9706, +0.0000, +81.9304, -0.0000, +182.5098, -0.0000, +317.3109, -0.0000, +471.8350, -0.0000, +574.4634, +0.0000, +612.5140, +0.0000, +514.8609, +0.0000, +384.0874, +0.0000, +198.4450, +0.0000, +93.3348, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$

$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
(+1.1200, −1.5900)
(+0.8100, −1.5900)
(−0.2400, −1.2000)
(−0.2400, −1.0300)
(+1.1200, −1.2000)
(+0.8100, −1.0300)
(+1.1200, +1.5900)
(+0.8100, +1.5900)
... (4 more eval-pairs)
PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(2, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +6.7600, +28.9706, +88.4413, +218.6210, +440.5627, +747.9918, +1056.1305, +1254.6568, +1207.9684, +922.4338, +489.6353, +157.4689
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
(−0.2400, −1.5900)
(−0.2400, −1.5900)
(+1.1200, −1.0300)
(+1.1200, −1.0300)
(+0.8100, −1.2000)
(+0.8100, −1.2000)
(−0.2400, +1.5900)
(−0.2400, +1.5900)
... (4 more eval-pairs)

### 3.56 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(2, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +3.3800, +4.9844, +3.8602, +2.3357, +1.5450, +0.8403
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$

$(\mathbf{a}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
Eval-pair multiset $(u, v)$:
$(-0.2400,\ -0.6430)$
$(+1.1200,\ -0.6430)$
$(+0.8100,\ -0.6430)$
$(+1.1200,\ +0.6430)$
$(+0.8100,\ +0.6430)$
$(-0.2400,\ +0.6430)$

### 3.57 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +3.6770, +7.2066, +10.3451, +9.7461, +6.3330, +2.7526
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a} \cdot \mathbf{b},\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{a} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{a} \cdot \mathbf{b},\ \mathbf{b} \cdot \mathbf{c})$
$(\mathbf{b} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{a} \cdot \mathbf{c},\ \mathbf{b} \cdot \mathbf{c})$
$(\mathbf{b} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{c})$
Eval-pair multiset $(u, v)$:
$(+0.9800,\ -0.5900)$
$(+0.9100,\ -0.5900)$
$(+0.9800,\ +0.9100)$
$(+0.9100,\ +0.9800)$
$(-0.5900,\ +0.9800)$
$(-0.5900,\ +0.9100)$
PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2, 0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a} \cdot \mathbf{b},\ \mathbf{a} \cdot \mathbf{b})$
$(\mathbf{a} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{c})$
$(\mathbf{b} \cdot \mathbf{c},\ \mathbf{b} \cdot \mathbf{c})$
Eval-pair multiset $(u, v)$:
$(-0.5900,\ -0.5900)$
$(+0.9100,\ +0.9100)$
$(+0.9800,\ +0.9800)$

### 3.58 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +2.6000, +1.2434, -1.6329, -1.3182, +0.2350, +0.2768
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))$
$(\mathbf{b} \cdot \mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
$(\mathbf{a} \cdot \mathbf{b},\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
$(\mathbf{a} \cdot \mathbf{c},\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))$
$(\mathbf{b} \cdot \mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$

Eval-pair multiset $(u, v)$:
$(-0.5900,\ +0.0000)$
$(-0.5900,\ +0.0000)$
$(+0.9800,\ +0.0000)$
$(+0.9800,\ -0.0000)$
$(+0.9100,\ -0.0000)$
$(+0.9100,\ -0.0000)$
PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +5.2000, +9.2468, +3.1999, -9.5814, -10.4452, +1.1640, +6.3288, +1.6586, -1.5236, -0.6746, +0.1301, +0.0766
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{c},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{c},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(-0.5900,\ +0.0000)$
$(-0.5900,\ +0.0000)$
$(-0.5900,\ +0.0000)$
$(-0.5900,\ -0.0000)$
$(+0.9800,\ -0.0000)$
$(+0.9800,\ -0.0000)$
$(+0.9800,\ +0.0000)$
$(+0.9800,\ +0.0000)$
. . . (4 more eval-pairs)

### 3.59 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +5.2000, +0.0000, +19.3048, -0.0000, +44.1431, -0.0000, +88.3113, -0.0000, +137.8122, -0.0000, +217.1767, -0.0000, +269.5218, -0.0000, +324.6213, -0.0000, +280.9677, -0.0000, +279.5834, -0.0000, +185.4809, -0.0000, +133.7534, -0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{b}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(-0.5900,\ +1.2000)$
$(-0.5900,\ +1.0300)$
$(+0.9800,\ +1.5900)$

$(+0.9100,\ +1.5900)$
$(+0.9800,\ +1.2000)$
$(+0.9100,\ +1.0300)$
$(-0.5900,\ -1.2000)$
$(-0.5900,\ -1.0300)$
... (4 more eval-pairs)
PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(2, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +5.2000, +19.3048, +52.0675, +125.1057, +249.4131, +435.0611, +636.0779, +824.2870, +888.3968, +786.4776, +473.5668, +173.8698
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\mathbf{b}\cdot\mathbf{c},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(\mathbf{b}\cdot\mathbf{c},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(-0.5900,\ -1.5900)$
$(-0.5900,\ -1.5900)$
$(-0.5900,\ +1.5900)$
$(-0.5900,\ +1.5900)$
$(+0.9100,\ -1.2000)$
$(+0.9100,\ -1.2000)$
$(+0.9800,\ -1.0300)$
$(+0.9800,\ -1.0300)$
... (4 more eval-pairs)

### 3.60 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(2, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +2.6000, +2.4837, +0.5170, +0.5921, +1.7446, +1.2990
Atom-pair orbit (multiset; ordering arbitrary):
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{b}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{a}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\mathbf{b}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
Eval-pair multiset $(u, v)$:
$(-0.5900,\ -0.6430)$
$(-0.5900,\ +0.6430)$
$(+0.9800,\ -0.6430)$
$(+0.9100,\ -0.6430)$
$(+0.9800,\ +0.6430)$
$(+0.9100,\ +0.6430)$

### 3.61 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=neg_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 2)$, type=`neg_sigma`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}),\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}),\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}),\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}),\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
. . . (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(−0.0000, −0.0000)
(−0.0000, −0.0000)
(−0.0000, −0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
. . . (4 more eval-pairs)
PairFlavour overlap=(1, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1, 2)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
$(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}),\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
$(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}),\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
$(-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}),\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
Eval-pair multiset $(u, v)$:
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(−0.0000, −0.0000)
(−0.0000, −0.0000)
(−0.0000, −0.0000)

### 3.62 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
$(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
$(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
$(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$

$(\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(-\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -1.5900)$
$(+0.0000,\ -1.5900)$
$(+0.0000,\ -1.2000)$
$(-0.0000,\ -1.2000)$
$(-0.0000,\ -1.0300)$
$(-0.0000,\ -1.0300)$
$(+0.0000,\ +1.5900)$
$(+0.0000,\ +1.5900)$
... (4 more eval-pairs)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, +10.0580, +0.0000, +0.0000, +0.0000, +40.9913, +0.0000, +0.0000, +0.0000, +86.6819, +0.0000, +0.0000, +0.0000, +100.4717, +0.0000, +0.0000, +0.0000, +60.6378, +0.0000, +0.0000, +0.0000, +14.9163, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ -1.5900)$
$(+0.0000,\ -1.5900)$
$(+0.0000,\ -1.2000)$
$(-0.0000,\ -1.2000)$
$(-0.0000,\ -1.0300)$
$(-0.0000,\ -1.0300)$
$(+0.0000,\ +1.5900)$
$(+0.0000,\ +1.5900)$
... (4 more eval-pairs)

### 3.63 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z}) \times \varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=22 kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`22`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: -2.4807, +2.5641, -1.4135, +0.4383, -0.0725, +0.0050
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$

$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+0.0000,\ +0.6430)$
$(+0.0000,\ +0.6430)$
$(+0.0000,\ -0.6430)$
$(-0.0000,\ -0.6430)$
$(-0.0000,\ +0.6430)$
$(-0.0000,\ +0.6430)$
$(+0.0000,\ -0.6430)$
$(+0.0000,\ -0.6430)$
... (4 more eval-pairs)

### 3.64 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z}) \times \varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, +0.0000, +9.2388, -0.0000, -0.0000, -14.7572, +0.0000, +0.0000, -0.0000, +0.0000, +98.5481, -0.0000, +0.0000, -301.4282, +0.0000, +0.0000, -0.0000, -0.0000, +257.2546, +0.0000, +0.0000, -1268.5418, -0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(-1.0300,\ +1.5900)$
$(-1.2000,\ +1.5900)$
$(-1.5900,\ +1.2000)$
$(-1.5900,\ +1.0300)$
$(+1.0300,\ +1.2000)$
$(+1.2000,\ +1.0300)$
$(-1.2000,\ -1.0300)$
$(-1.0300,\ -1.2000)$
... (4 more eval-pairs)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
**Encoded values**: +0.0000, +0.0000, +0.0000, -9.2388, +0.0000, -0.0000, -14.7572, -0.0000, +0.0000, +0.0000, -0.0000, -98.5481, +0.0000, -0.0000, -301.4282, -0.0000, -0.0000, -0.0000, -0.0000, -257.2546, +0.0000, +0.0000, -1268.5418, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$

$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+1.5900,\ +1.0300)$
$(+1.5900,\ +1.2000)$
$(+1.2000,\ +1.5900)$
$(+1.0300,\ +1.5900)$
$(+1.2000,\ -1.0300)$
$(+1.0300,\ -1.2000)$
$(-1.0300,\ +1.2000)$
$(-1.2000,\ +1.0300)$
... (4 more eval-pairs)
PairFlavour overlap=(2, 0) type=neg_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(2, 0)$, type=`neg_sigma`, kind=`ASSOC`, encoded dim= 6
**Encoded values**: -20.1160, -163.9651, +693.4549, +1607.5466, -1940.4080, -954.6460
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(-\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(-\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+1.5900,\ -1.5900)$
$(-1.5900,\ +1.5900)$
$(+1.5900,\ -1.5900)$
$(-1.5900,\ +1.5900)$
$(+1.2000,\ -1.2000)$
$(-1.2000,\ +1.2000)$
$(+1.2000,\ -1.2000)$
$(-1.2000,\ +1.2000)$
... (4 more eval-pairs)
PairFlavour overlap=(2, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+1.0300,\ +1.0300)$
$(+1.0300,\ +1.0300)$
$(+1.2000,\ +1.2000)$
$(+1.2000,\ +1.2000)$

$(+1.5900,\ +1.5900)$
$(+1.5900,\ +1.5900)$
$(-1.0300,\ -1.0300)$
$(-1.0300,\ -1.0300)$
... (4 more eval-pairs)

### 3.65 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=neg kind=ASSOC dim=12 **PairFlavour** overlap= $(2, 0)$, type=`neg`, kind=`ASSOC`, encoded dim= 12
**Encoded values**: +7.5773, +0.0000, +31.0800, -0.0000, +72.8464, -0.0000, +116.6920, -0.0000, +108.6872, -0.0000, +64.6121, -0.0000
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
... (4 more atom-pairs)
Eval-pair multiset $(u, v)$:
$(+1.5900,\ -0.6430)$
$(-1.5900,\ +0.6430)$
$(+1.5900,\ +0.6430)$
$(-1.5900,\ -0.6430)$
$(+1.2000,\ -0.6430)$
$(-1.2000,\ +0.6430)$
$(+1.0300,\ -0.6430)$
$(-1.0300,\ +0.6430)$
... (4 more eval-pairs)

### 3.66 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(3, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(3, 0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
Eval-pair multiset $(u, v)$:
$(+0.6430,\ +0.6430)$
$(-0.6430,\ -0.6430)$

## 4 Alignment Decoder (recovers the G-orbit of repS evaluations)

**—repS—**: $|repS| = 11$

**—G—**: $|G| = 12$

**Decoded vectors**: decoded column vectors: 12

Decoded alignment table (rows=repS atoms, cols=group elements): **Decoded alignment table** (rows= repS atoms, columns= $g \in G$):

Table 1: Decoded alignment table

| Atom | $g_0$ | $g_1$ | $g_2$ | $g_3$ | $g_4$ | $g_5$ | $g_6$ | $g_7$ | $g_8$ | |
|---|---|---|---|---|---|---|---|---|---|---|
| $\|\mathbf{a}\|$ | +1.4765 | +1.4765 | +1.3342 | +1.3342 | +1.1874 | +1.1874 | +1.1874 | +1.1874 | +1.3342 | +1.3 |
| $\|\mathbf{p}\|$ | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0 |
| $\mathbf{a} \cdot \mathbf{b}$ | -0.5900 | +0.9800 | -0.5900 | +0.9100 | +0.9800 | +0.9100 | +0.9100 | +0.9800 | +0.9100 | -0.5 |
| $\mathbf{a} \cdot \mathbf{p}$ | -0.7000 | -0.7000 | -0.5000 | -0.5000 | -0.2000 | -0.2000 | +0.2000 | +0.2000 | +0.5000 | +0.5 |
| $\mathbf{p} \cdot \mathbf{q}$ | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0 |
| $\mathbf{a}_T \cdot \mathbf{b}_T$ | -0.2400 | +1.1200 | -0.2400 | +0.8100 | +1.1200 | +0.8100 | +0.8100 | +1.1200 | +0.8100 | -0.2 |
| $\mathbf{a}_T \cdot \mathbf{p}_T$ | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0 |
| $\mathbf{p}_T \cdot \mathbf{q}_T$ | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0 |
| $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$ | -0.6430 | -0.6430 | -0.6430 | -0.6430 | -0.6430 | -0.6430 | -0.6430 | -0.6430 | -0.6430 | -0.6 |
| $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$ | -1.5900 | -1.0300 | -1.5900 | -1.2000 | -1.0300 | +1.2000 | -1.2000 | +1.0300 | +1.2000 | +1.5 |
| $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$ | +0.0000 | +0.0000 | +0.0000 | +0.0000 | -0.0000 | -0.0000 | -0.0000 | -0.0000 | -0.0000 | -0.0 |

Ground truth (direct evaluation of g·repS on event): **Ground truth** (direct evaluation of $g \cdot$ repS on event):

Table 2: Ground truth alignment table

| Atom | $g_0$ | $g_1$ | $g_2$ | $g_3$ | $g_4$ | $g_5$ | $g_6$ | $g_7$ | $g_8$ | |
|---|---|---|---|---|---|---|---|---|---|---|
| $\|\mathbf{a}\|$ | +1.3342 | +1.3342 | +1.3342 | +1.3342 | +1.1874 | +1.1874 | +1.1874 | +1.1874 | +1.4765 | +1.4 |
| $\|\mathbf{p}\|$ | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0000 | +1.0 |
| $\mathbf{a} \cdot \mathbf{b}$ | +0.9100 | +0.9100 | -0.5900 | -0.5900 | +0.9100 | +0.9100 | +0.9800 | +0.9800 | -0.5900 | -0.5 |
| $\mathbf{a} \cdot \mathbf{p}$ | +0.5000 | -0.5000 | +0.5000 | -0.5000 | +0.2000 | -0.2000 | +0.2000 | -0.2000 | -0.7000 | +0.7 |
| $\mathbf{p} \cdot \mathbf{q}$ | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0000 | -1.0 |
| $\mathbf{a}_T \cdot \mathbf{b}_T$ | +0.8100 | +0.8100 | -0.2400 | -0.2400 | +0.8100 | +0.8100 | +1.1200 | +1.1200 | -0.2400 | -0.2 |
| $\mathbf{a}_T \cdot \mathbf{p}_T$ | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0 |
| $\mathbf{p}_T \cdot \mathbf{q}_T$ | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0000 | +0.0 |
| $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$ | -0.6430 | -0.6430 | +0.6430 | +0.6430 | +0.6430 | +0.6430 | -0.6430 | -0.6430 | -0.6430 | -0.6 |
| $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$ | +1.2000 | -1.2000 | +1.5900 | -1.5900 | -1.2000 | +1.2000 | +1.0300 | -1.0300 | -1.5900 | +1.5 |
| $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$ | +0.0000 | -0.0000 | +0.0000 | -0.0000 | +0.0000 | -0.0000 | +0.0000 | -0.0000 | +0.0000 | -0.0 |

Multiset comparison (decoded vs ground truth): **Multiset comparison:**
**MISMATCH** — decoded multiset differs from ground truth

Max —error— per repS row (decoded vs ground truth): **Max |error| per repS row:**
**Mymag(a)**: $|\mathbf{a}|$: max err $= 0.00e + 00$

**Mymag(p)**: $|\mathbf{p}|$: max err $= 0.00e + 00$

**dot(a,b)**: $\mathbf{a} \cdot \mathbf{b}$: max err $= 0.00e + 00$

**dot(a,p)**: $\mathbf{a} \cdot \mathbf{p}$: max err $= 0.00e + 00$

**dot(p,q)**: $\mathbf{p} \cdot \mathbf{q}$: max err $= 0.00e + 00$

**dot(a,b)**: $\mathbf{a}_T \cdot \mathbf{b}_T$: max err $= 0.00e + 00$

**dot(a,p)**: $\mathbf{a}_T \cdot \mathbf{p}_T$: max err $= 0.00e + 00$

**dot(p,q)**: $\mathbf{p}_T \cdot \mathbf{q}_T$: max err $= 0.00e + 00$

**eps(a,b,c)**: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$: max err $= 1.29e + 00$

**eps(a,b,p)**: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$: max err $= 0.00e + 00$

**eps(a,p,q)**: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$: max err $= 0.00e + 00$

## 5 Summary

**Phase 1 encoded length**: Phase 1 encoded length: 35 reals

**Phase 2 encoded length**: Phase 2 encoded length: 1214 reals

**Total encoded length**: Total encoded length: 1249 reals

**repS size**: |repS| = 11 atoms

**—G—**: $|G| = 12$ group elements

**Decoded vectors**: decoded alignment vectors: 12

TeX output written to: /Users/lester/github/Echinocoder/symcoder/demo_roundtrip.tex Compile with: pdflatex demo_roundtrip.tex